

Multiplication Masters

Every graded set, every week — 5 weeks, 25 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Arrays Become Area

Warm-Up facts, then break each big number into tens and ones.

Warm-Up 12 ITEMS • RECALL

1 6×7

2 8×4

3 9×6

4 7×7

5 2×7

6 5×6

7 3×4

8 2×9

9 5×9

10 8×8

11 12×3

12 11×5

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

13. 4×13 ($40 + 12$)

14. 6×14 ($60 + 24$)

15. 3×27 ($60 + 21$)

16. 5×18 ($50 + 40$)

17. 7×16 ($70 + 42$)

18. Area 84, one side 6. Other side?

★

Challenge

NOT GRADED · REASON IT OUT

19. How many rectangles have area 36?
(84 ÷ 6)

.....

20. 10 × 8
(1×36, 2×18, 3×12, 4×9, 6×6)

.....

21. 12 × 6

.....

22. 11 × 9

.....

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Breaking Apart

One-digit times two-digit. Two rooms every time.

Warm-Up 12 ITEMS • RECALL

1 5×9

2 8×8

3 7×8

4 6×9

5 3×6

6 4×5

7 2×8

8 5×5

9 3×9

10 4×9

11 7×6

12 9×9

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $4 \times 12(240 + 32)$

14. $3 \times 11(240 + 42)$

15. $8 \times 34(180 + 54)$

16. $6 \times 47(350 + 56)$

17. $9 \times 26(360 + 24)$

18. 7×58

19. 4×96

20. $6 \times 4\square = 2\blacktriangle 6$ — what is $\square$?



Challenge

21. $4 \times 68 = ?$ ($6 \times 46 = 276$)

22. 10×7 (Same as 8×34)

23. 12×8

24. 7×12



Working space



Error journal

Mental Math Moves

Go past the number, then take back the extra. No pencil.

Warm-Up 12 ITEMS • RECALL

1 10×12

2 9×6 — go to 10×6 , then take one 6 back

3 5×8 — half of 10×8

4 6×9 — go to 6×10 , then take one 6 back

5 25×4

6 25×8

7 50×6

8 20×7

9 15×4

10 30×9

11 19×6

12 99×7

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $48 \times 5 (20 \times 6 - 6)$

14. $25 \times 12 (700 - 7)$

15. $102 \times 4 (\text{Half of } 480)$

16. $35 \times 6 (100 \times 3)$

17. $998 \times 5 (400 + 8)$

18. $16 \times 25 (180 + 30)$

19. 20×3

20. 30×2

21. 40×2

22. 50×3

23. 60×2

The Four Rooms

Two 2-digit numbers make four rooms. Count them before you add.

Warm-Up 12 ITEMS • RECALL

1 10×10

2 20×30

3 40×50

4 12×10

5 60×70

6 90×80

7 7×8 as two rooms of 7×4 — the total

8 A rectangle 6 by 9 — the area

9 Area 72, one side 8 — the other side

10 4×30

11 6×40

12 23×14

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 36×25 ($200 + 80 + 30 + 12$)
.....

14. 47×32 ($600 + 150 + 120 + 30$)
.....

15. 58×46 ($1200 + 210 + 80 + 14$)
.....

16. 64×27 ($2000 + 320 + 300 + 48$)
.....

17. In 23×14 , which two rooms make 92? Type their sum. ($1200 + 420 + 80 + 28$)
.....

18. $25 \times 25 = 625$. What is 26×26 ?
.....

19. 11×2
.....

20. 12×3
.....

21. 13×2
.....

22. 20×4
.....

★ Challenge NOT GRADED • REASON IT OUT

23. $15 \times 2(80 + 12)$

.....

24. $14 \times 2(625 + 25 + 25 + 1)$

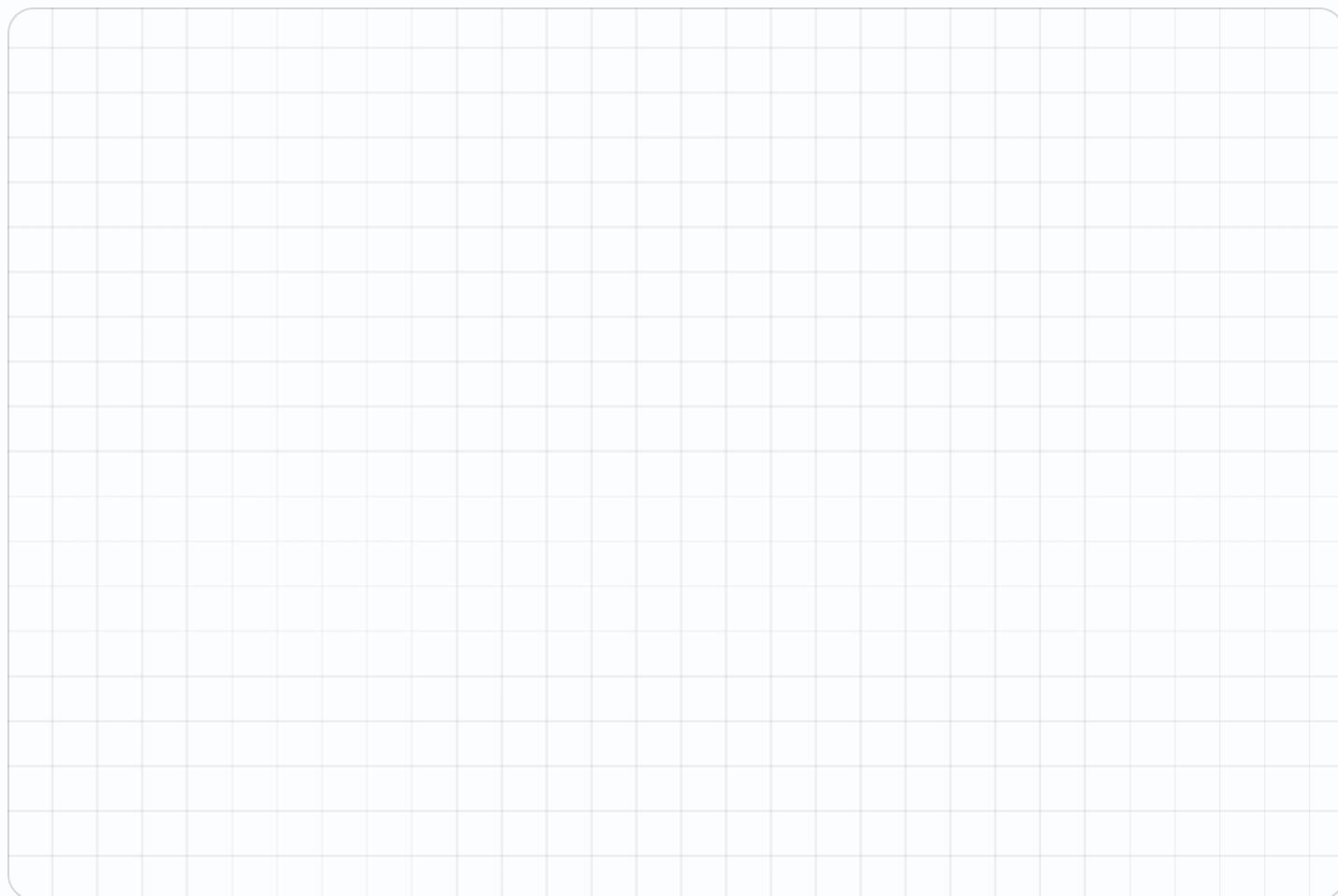
.....

25. 21×14

.....

26. 32×12

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Rectangle Hunt & Puzzles

Enrichment day. Reason it out — no guessing.

✦ Warm-Up 2 ITEMS • RECALL

1 4×4

2 6×6

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

3. 9×7

4. 8×6



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • Arrays Become Area

OUT OF 10

Warm-Up • 1-9 • 42 | 32 | 54 | 49 | 14 | 30 | 12 | 18 | 45

Core • 10-10 • 64

Challenge • 11-22 • 36 | 55 | 52 | 84 | 81 | 90 | 112 | 14 | 5 | 80 | 72 | 99

1.2 • Breaking Apart

OUT OF 12

Warm-Up • 1-9 • 45 | 64 | 56 | 54 | 18 | 20 | 16 | 25 | 27

Core • 10-12 • 36 | 42 | 81

Challenge • 13-24 • 48 | 33 | 272 | 282 | 234 | 406 | 384 | 6 | 272 | 70 | 96 | 84

1.3 • Mental Math Moves

OUT OF 4

Warm-Up • 1-1 • 120

Core • 2-4 • 54 | 40 | 54

Challenge • 5-30 • 100 | 200 | 300 | 140 | 60 | 270 | 114 | 693 | 240 | 300 | 408
| 210 | 4990 | 400 | 60 | 60 | 80 | 150 | 120 | 69 | 68 | 168 | 306 | 189 |
496 | 645

1.4 • The Four Rooms

OUT OF 11

Warm-Up • 1-6 • 100 | 600 | 2000 | 120 | 4200 | 7200

Core • 7-11 • 56 | 54 | 9 | 120 | 240

Challenge • 12-31 • 322 | 900 | 1504 | 2668 | 1728 | 92 | 676 | 22 | 36 | 26 | 80
| 30 | 28 | 294 | 384 | 400 | 559 | 270 | 972 | 1680

Fri • Rectangle Hunt & Puzzles

OUT OF 4

Warm-Up • 1-2 • 16 | 36

Core • 3-4 • 63 | 48

Challenge • 5-15 • 6 | 3 | 6 | 6 | 44 | 5 | 1 | 5 | 100 | 180 | 297

Where the Rooms Hide

Every line of the algorithm is one of the four rooms. Find it before you trust it.

Warm-Up 12 ITEMS • RECALL

1 20×10

2 3×4

3 Add 200, 30, 80 and 12

4 So 23×14 is

5 8×7 as $8 \times 5 + 8 \times 2$ —
the total

6 The 8×5 room of 8×7

7 3×10

8 20×4

9 31×22 — the tensxtens
room

10 31×22 — the
onesxones room

11 31×22 altogether

12 45×23

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 18×26 (30×20)
.....

14. In 47×35 , which room is biggest — type its value
.....

15. 47×35
.....

16. 2×20
.....

17. 3×30
.....

18. 4×20
.....

19. 5×40
.....

20. 6×10
.....

21. 7×20
.....

22. 8×30
.....

Fluency with Carrying

Algorithm only. Use the area model to check, not to solve.

Warm-Up 12 ITEMS • RECALL

1 14×12

2 21×13

3 32×11

4 15×15

5 24×12

6 13×13

7 7×6

8 9×8

9 6×6

10 5×70

11 8×40

12 38×24

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 56×27

14. 49×36

15. 63×48

16. 75×29

17. 87×64

18. 99×99

19. 2×13

20. 3×12

21. 4×11

22. 2×24

Three Digits by One

A longer rectangle, split into three rooms instead of two.

Warm-Up 12 ITEMS • RECALL

1 9×8

2 10×10

3 20×20

4 30×10

5 40×30

6 50×20

7 60×30

8 3×24 as $3 \times 20 + 3 \times 4$

9 The 3×20 room of 3×24

10 2×43 as $2 \times 40 + 2 \times 3$

11 7×60

12 9×30

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $100 \times 4 (600 + 90 + 12)$

14. 200×3

15. 30×7

16. 400×2

17. 50×6

18. 234×3

19. 417×5

20. 326×4

21. 508×7

22. 739×6

Estimate First

Round, predict, compute, compare. A wrong answer should look wrong.

Warm-Up 12 ITEMS • RECALL

1 Estimate 19×21 by rounding both

2 Estimate 29×31

3 Estimate 48×12

4 Estimate 39×41

5 Estimate 22×18

6 Round 48 to the nearest ten

7 Estimate 4×19 by rounding 19 to 20

8 4×19 — the true value

9 Estimate 3×31 by rounding 31 to 30

10 3×31 — the true value

11 Estimate 51×9

12 Estimate 187×4

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. True value of 187×4

.....

14. Estimate 62×38

.....

15. True value of 62×38

.....

16. Estimate 296×5

.....

17. Someone says $43 \times 27 = 301$. Type the real answer

.....

18. Estimate 412×19

.....

19. 5×12

.....

20. 6×11

.....

21. 4×25

.....

22. 2×50

.....

Lattice Detour

Enrichment. A 500-year-old method — then decide whether it beats yours.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

 Warm-Up 4 ITEMS • RECALL

1 6×7

2 Digits in the answer to 23×14

3 7×5

4 9×4

 Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Lattice cells needed for 2 digits $\times$ 2 digits
.....
6. Lattice cells for 3 digits $\times$ 2 digits
.....
7. Rows of 6 with 7 in each — how many
.....
8. Area 45, one side 5 — the other side
.....

★ Challenge NOT GRADED • REASON IT OUT9. 23×14 by lattice — the answer10. 456×23 — the answer

11. Partial products in a 3-by-2 lattice

12. 12×12 13. 16×15 14. 35×24 **✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Where the Rooms Hide

OUT OF 6

Warm-Up • 1-4 • 200 | 12 | 322 | 322

Core • 5-6 • 56 | 40

Challenge • 7-28 • 30 | 80 | 600 | 2 | 682 | 1035 | 468 | 1200 | 1645 | 40 | 90 | 80 | 200 | 60 | 140 | 240 | 360 | 420 | 420 | 720 | 480 | 750

2.2 • Fluency with Carrying

OUT OF 11

Warm-Up • 1-6 • 168 | 273 | 352 | 225 | 288 | 169

Core • 7-11 • 42 | 72 | 36 | 350 | 320

Challenge • 12-31 • 912 | 1512 | 1764 | 3024 | 2175 | 5568 | 9801 | 26 | 36 | 44 | 48 | 63 | 55 | 144 | 224 | 184 | 369 | 270 | 896 | 2106

2.3 • Three Digits by One

OUT OF 12

Warm-Up • 1-7 • 72 | 100 | 400 | 300 | 1200 | 1000 | 1800

Core • 8-12 • 72 | 60 | 86 | 420 | 270

Challenge • 13-31 • 400 | 600 | 210 | 800 | 300 | 702 | 2085 | 1304 | 3556 | 4434 | 9960 | 457 | 294 | 736 | 744 | 966 | 1612 | 1786 | 3648

2.4 • Estimate First

OUT OF 10

Warm-Up • 1-5 • 400 | 900 | 500 | 1600 | 400

Core • 6-10 • 50 | 80 | 76 | 90 | 93

Challenge • 11-31 • 500 | 800 | 748 | 2400 | 2356 | 1500 | 1161 | 8000 | 60 | 66 | 100 | 100 | 80 | 99 | 210 | 338 | 748 | 540 | 609 | 2408 | 2028

Fri • Lattice Detour

OUT OF 8

Warm-Up • 1-4 • 42 | 3 | 35 | 36

Core • 5-8 • 4 | 6 | 144 | 240

Challenge • 9-14 • 322 | 10488 | 6 | 840 | 936 | 1000

Factor Pairs

Every rectangle you can build is a factor pair.

Warm-Up 12 ITEMS • RECALL

1 Factor pairs of 12 — how many

2 Smallest factor of any number

3 $12 \div 3$

4 Is 5 a factor of 20 — yes or no

5 Is 7 a factor of 20

6 Largest factor of 20

7 6×8

8 7×9

9 Factor pairs of 24 — how many

10 Factor pairs of 36 — how many

11 All factors of 18 — how many

12 Factors of 30 — how many

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. The missing partner of 4 in 48
.....

14. A number under 30 with exactly 8 factors
.....

15. Highest factor of 48 below 48
.....

16. 4×12
.....

17. 5×14
.....

18. 3×15
.....

19. 2×45
.....

20. 18×6
.....

21. 24×7
.....

22. 36×5
.....

Factor Rainbows

Draw the arcs. Where they stop is the question worth asking.

Warm-Up 12 ITEMS • RECALL

1 Factors of 16 — how many

2 Partner of 2 in 16

3 Partner of 1 in 16

4 Middle factor of 16

5 Factors of 9 — how many

6 Middle factor of 9

7 Rows of 4, 3 rows — total

8 Rows of 5, 4 rows

9 Rows of 6, 3 rows

10 Rows of 10, 7 rows

11 Rows of 2, 9 rows

12 Rows of 8, 2 rows

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Highest number you must test for factors of 36 (Its square root)

14. Highest you must test for 100

15. Factors of 100 — how many

16. Arcs in the rainbow of 24

17. Factors of 49 — how many

18. Rows of 12, 8 rows

19. Rows of 15, 6 rows

20. Rows of 14, 7 rows

21. Rows of 24, 5 rows

22. Rows of 16, 9 rows



Common Factors

Two numbers, the factors they share, and the biggest of those.

Warm-Up 12 ITEMS • RECALL

1 A factor shared by every pair of numbers

2 Common factors of 6 and 9 — the greatest

3 Greatest common factor of 4 and 8

4 GCF of 5 and 10

5 GCF of 3 and 7

6 GCF of 12 and 12

7 2×6

8 6×2

9 3×7

10 7×3

11 4×8

12 8×4

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. GCF of 12 and 18

.....

14. GCF of 24 and 36

.....

15. GCF of 15 and 25

.....

16. GCF of 16 and 40

.....

17. GCF of 27 and 45

.....

18. $(2 \times 3) \times 5$

.....

19. $2 \times (3 \times 5)$

.....

20. $5 \times 7 \times 2$

.....

21. $8 \times 5 \times 3$

.....

22. GCF of 48 and 72

.....



Challenge

NOT GRADED • REASON IT OUT

23. Two numbers whose only common factor is 1 are called

24. $4 \times 25 \times 2$

25. $2 \times 3 \times 4 \times 5$

26. $25 \times 4 \times 11$



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal

ONE SENTENCE PER MISTAKE

Square Numbers

Why do squares have an odd number of factors? Build the rectangles and see.

Warm-Up 12 ITEMS • RECALL

1 3×3

2 5×5

3 7×7

4 10×10

5 4×4

6 8×8

7 9×2

8 9×3

9 9×5

10 9×7

11 9×8

12 Factors of 25 — how many

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Factors of 36 — how many
.....14. The square number between 60 and 70
.....15. The square number between 40 and 50
.....16. A square with side 6 — the area
.....17. 12×12
.....18. 15×15
.....19. Square numbers between 1 and 100 — how many
.....20. A square's factor count is always odd or even — type one
.....21. 9×10
.....22. 9×12
.....

★ Challenge NOT GRADED • REASON IT OUT

23. 9×15

.....

24. 11×7

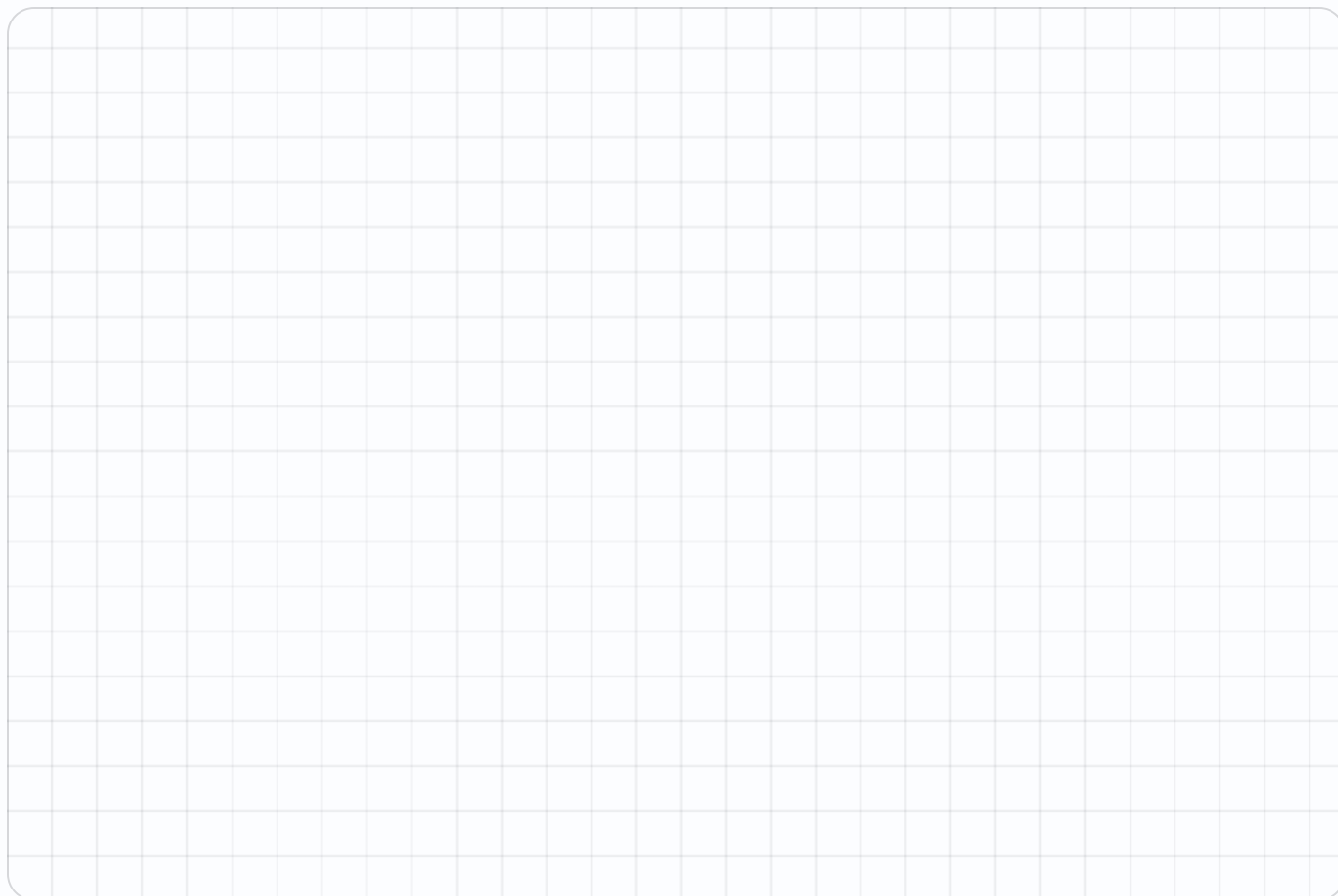
.....

25. 11×12

.....

26. 12×9

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 7×8

2 8×3

3 7×4

4 Factor pairs of 24 — how many

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. GCF of 12 and 18
.....

6. 8×7
.....

7. A rectangle 7 by 9 — the area
.....

8. 6×40
.....

9. 8×34
.....

10. 23×14
.....

11. Estimate 39×21
.....



Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Factor Pairs

OUT OF 13

Warm-Up • 1-8 • 3 | 1 | 4 | yes | no | 20 | 48 | 63

Core • 9-13 • 4 | 5 | 6 | 8 | 12

Challenge • 14-26 • 24 | 24 | 48 | 70 | 45 | 90 | 108 | 168 | 180 | 192 | 216 | 864 | 1652

3.2 • Factor Rainbows

OUT OF 22

Warm-Up • 1-12 • 5 | 8 | 16 | 4 | 3 | 3 | 12 | 20 | 18 | 70 | 18 | 16

Core • 13-22 • 6 | 10 | 9 | 4 | 3 | 96 | 90 | 98 | 120 | 144

Challenge • 23-26 • square | 15 | 600 | 540

3.3 • Common Factors

OUT OF 21

Warm-Up • 1-12 • 1 | 3 | 4 | 5 | 1 | 12 | 12 | 12 | 21 | 21 | 32 | 32

Core • 13-21 • 6 | 12 | 5 | 8 | 9 | 30 | 30 | 70 | 120

Challenge • 22-26 • 24 | coprime | 200 | 120 | 1100

3.4 • Square Numbers

OUT OF 16

Warm-Up • 1-11 • 9 | 25 | 49 | 100 | 16 | 64 | 18 | 27 | 45 | 63 | 72

Core • 12-16 • 3 | 9 | 64 | 49 | 36

Challenge • 17-28 • 144 | 225 | 10 | odd | 90 | 108 | 135 | 77 | 132 | 108 | 693 | 3996

Fri • Mid-Unit Quiz

OUT OF 11

Warm-Up • 1-4 • 56 | 240 | 24 | 28

Core • 5-11 • 272 | 322 | 4 | 6 | 800 | 84 | 207

Challenge • 12-15 • 14 | 1440 | 1407 | 1000

Multiples

Skip-count on the 100-grid and watch where the patterns cross.

Warm-Up 12 ITEMS • RECALL

1 Third multiple of 4

2 Fifth multiple of 3

3 Fourth multiple of 6

4 Is 21 a multiple of 3 — yes or no

5 Is 22 a multiple of 3

6 3×3

7 5×8

8 6×4

9 4×7

10 First number that is a multiple of both 4 and 6

11 First multiple of both 3 and 5

12 First multiple of both 6 and 8

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Multiples of 7 below 50 — how many

.....

14. First multiple of both 4 and 10

.....

15. Tenth multiple of 5

.....

16. Lowest common multiple of 9 and 12

.....

17. Numbers below 100 that are multiples of both 3 and 4

.....

18. 2×12

.....

19. 10×9

.....

20. 13×7

.....

21. 17×6

.....

22. 19×8

.....



Challenge

23. 22×9

24. 31×7

25. 123×8

26. 307×6



Working space



Error journal

Divisibility Rules

Tests for 2, 3, 5, 9 and 10 — and why the 3-rule works at all.

Warm-Up 12 ITEMS • RECALL

1 Is 48 divisible by 2 — yes or no

2 Is 35 divisible by 5

3 Is 47 divisible by 2

4 Is 90 divisible by 10

5 Digit sum of 51

6 Is 51 divisible by 3

7 \$2 each, 4 items

8 \$3 each, 5 items

9 \$5 each, 6 items

10 \$10 each, 3 items

11 \$4 each, 4 items

12 \$6 each, 2 items

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Digit sum of 738
.....

14. Is 738 divisible by 9 — yes or no
.....

15. Is 4,671 divisible by 3
.....

16. Is 1,234 divisible by 3
.....

17. Smallest digit that makes 12_ divisible by 3
.....

18. \$12 each, 7 items
.....

19. \$15 each, 6 items
.....

20. \$24 each, 5 items
.....

21. \$18 each, 9 items
.....

22. \$35 each, 4 items
.....



Working space



Error journal

Primes & Composites

Colour the sieve. Every prime below 100 on one page.

Warm-Up 12 ITEMS • RECALL

1 Is 7 prime — yes or no

2 Is 9 prime

3 Is 2 prime

4 Is 1 prime

5 Factors a prime has

6 The only even prime

7 Primes below 20 — how many

8 Primes below 50 — how many

9 The prime just after 23

10 Is 51 prime — yes or no

11 The largest prime below 100

12 12×11

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 14×11

.....

14. 23×11

.....

15. 35×11

.....

16. $42 \times 11 (3 \times 17)$

.....

17. Primes below 100 — how many

.....

18. Is 91 prime — yes or no

.....

19. 2×11

.....

20. 3×11

.....

21. 4×11

.....

22. 5×11

.....

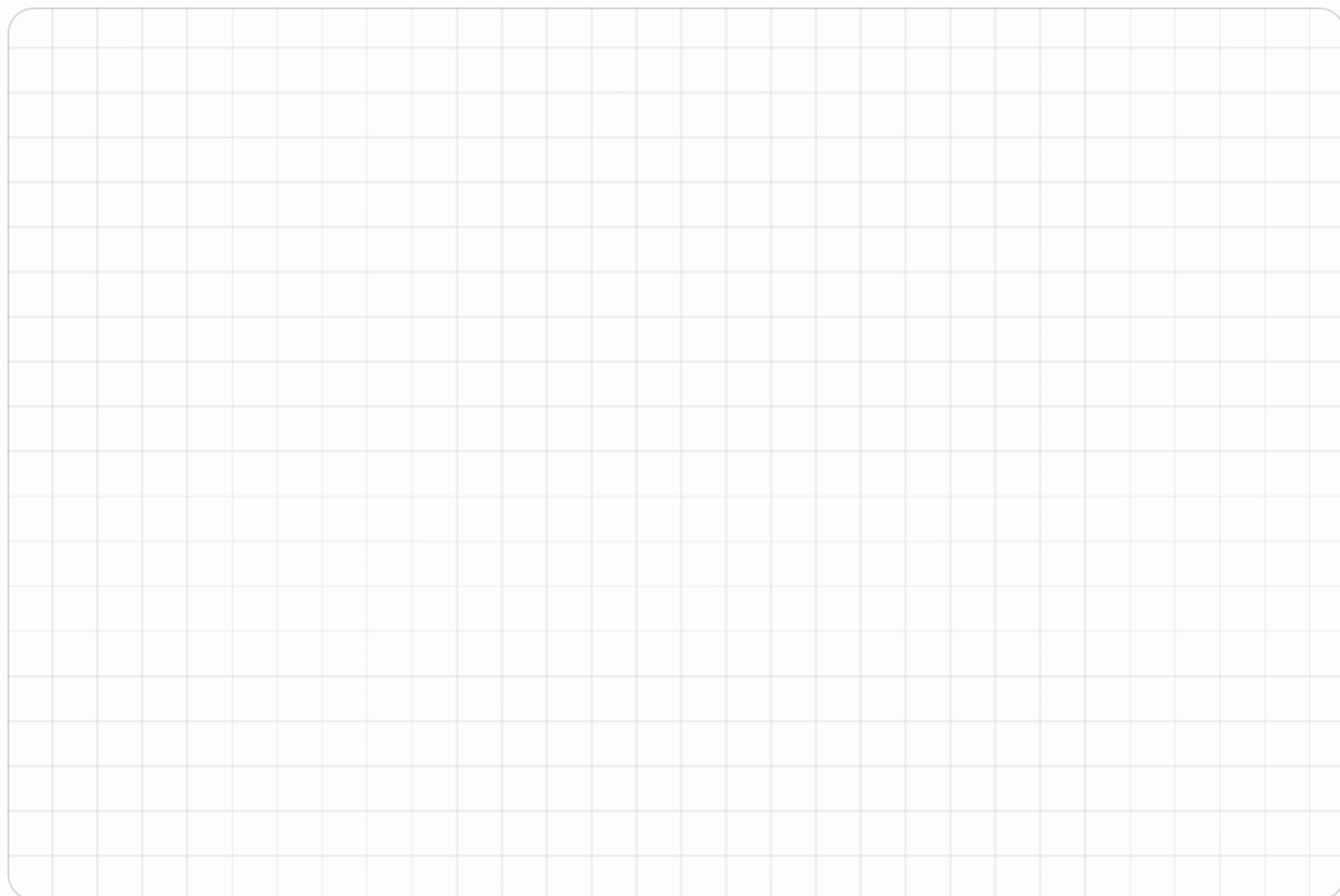
★ Challenge NOT GRADED • REASON IT OUT

23. 6×11
.....

24. 7×11 (7×13)
.....

25. 87×11
.....

26. 123×11
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Prime Factor Trees

Break a number all the way down until only primes are left.

Warm-Up 12 ITEMS • RECALL

1 12 as $2 \times ?$

2 6 as $2 \times ?$

3 Prime factors of 12 —
how many, counting repeats

4 8 as $2 \times 2 \times ?$

5 Prime factors of 10 — the
larger one

6 Prime factors of 15 — the
smaller

7 5×7

8 8×9

9 6×5

10 9×4

11 7×2

12 8×5

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Prime factors of 36 counting repeats ($2 \times 2 \times 3 \times 3$)

.....

14. Largest prime factor of 60

.....

15. Largest prime factor of 84

.....

16. Prime factors of 100 counting repeats

.....

17. Largest prime factor of 45

.....

18. 16×12

.....

19. 25×14

.....

20. 33×12

.....

21. 27×15

.....

22. 44×13

.....



Working space



Error journal

Twin Primes

Enrichment. Pairs two apart — and nobody knows whether they ever stop.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

✦

Warm-Up

4 ITEMS • RECALL

- 1

The prime two above 3

- 2

The prime two above 5

- 3

6×3

- 4

9×5

◆

Core

GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5.

The prime two above 11

.....
6.

The prime two above 17

.....
7.

26×12

.....
8.

Twin prime pairs below 50 — how many

.....

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Multiples

OUT OF 14

Warm-Up • 1-9 • 12 | 15 | 24 | yes | no | 9 | 40 | 24 | 28

Core • 10-14 • 12 | 15 | 24 | 7 | 20

Challenge • 15-26 • 50 | 36 | 8 | 24 | 90 | 91 | 102 | 152 | 198 | 217 | 984 | 1842

4.2 • Divisibility Rules

OUT OF 22

Warm-Up • 1-12 • yes | yes | no | yes | 6 | yes | 8 | 15 | 30 | 30 | 16 | 12

Core • 13-22 • 18 | yes | yes | no | 0 | 84 | 90 | 120 | 162 | 140

Challenge • 23-26 • 6 | yes | 1000 | 564

4.3 • Primes & Composites

OUT OF 16

Warm-Up • 1-6 • yes | no | yes | no | 2 | 2

Core • 7-16 • 8 | 15 | 29 | no | 97 | 132 | 154 | 253 | 385 | 462

Challenge • 17-26 • 25 | no | 22 | 33 | 44 | 55 | 66 | 77 | 957 | 1353

4.4 • Prime Factor Trees

OUT OF 22

Warm-Up • 1-12 • 6 | 3 | 3 | 2 | 5 | 3 | 35 | 72 | 30 | 36 | 14 | 40

Core • 13-22 • 4 | 5 | 7 | 4 | 5 | 192 | 350 | 396 | 405 | 572

Challenge • 23-26 • 7 | 60 | 2668 | 3367

Fri • Twin Primes

OUT OF 8

Warm-Up • 1-4 • 5 | 7 | 18 | 45

Core • 5-8 • 13 | 19 | 162 | 312

Challenge • 9-14 • 6 | 31 | 2 | 1600 | 3066 | 3000

Missing-Digit Puzzles

Reason backwards from the product. The last digit gives it away.

Warm-Up 12 ITEMS • RECALL

1 $4 \times ? = 24$

2 $? \times 7 = 56$

3 $9 \times ? = 81$

4 $? \times 6 = 42$

5 $12 \times ? = 60$

6 $? \times 11 = 88$

7 4×6

8 3×8

9 5×4

10 9×3

11 $3 \times 4? = 129$ — the missing digit

12 $? \times 14 = 322$ — the number

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $6 \times 4? = 276$ — the missing digit
.....

14. $2? \times 3 = 84$ — the missing digit
.....

15. $? \times 12 = 156$
.....

16. $?? \times ?? = 1,024$ using two equal 2-digit numbers — type one
.....

17. $4? \times 5$ ends in 5. The missing digit can be 1, 3, 5, 7 or 9 — type the smallest
.....

18. 7×10
.....

19. 2×25
.....

20. 21×8
.....

21. 34×6
.....

22. 47×5
.....

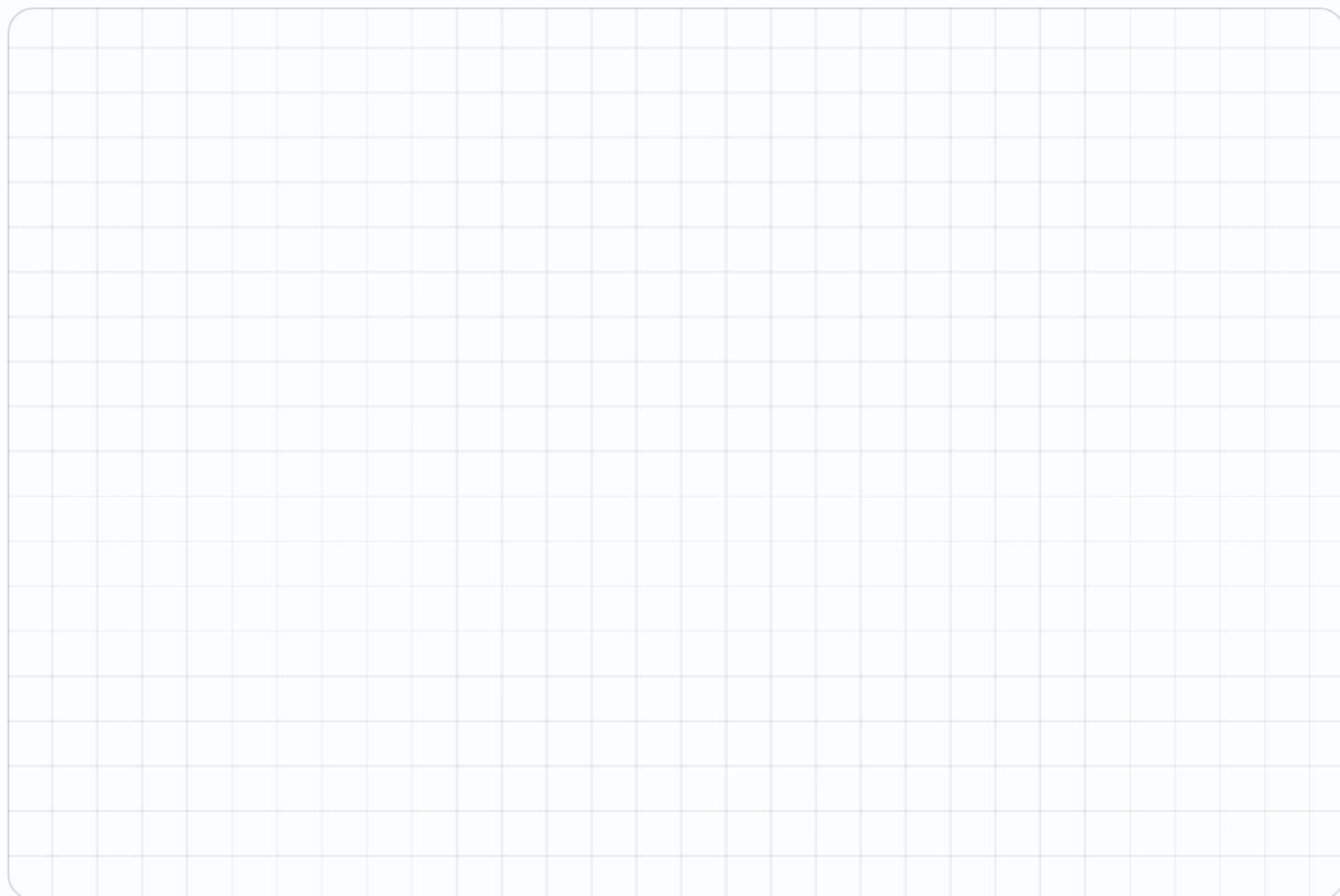
★ Challenge NOT GRADED • REASON IT OUT

23. 56×7
.....

24. 38×9
.....

25. 216×9
.....

26. 408×7
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Launch Bay Blueprint

Design the hangar, then find its area two different ways.

Warm-Up 12 ITEMS • RECALL

1 Area of a 10 by 8 bay

2 Area of a 12 by 5 bay

3 Perimeter of a 10 by 8 bay

4 Area of a 6 by 6 bay

5 Area of a 20 by 3 bay

6 Perimeter of a 6 by 6 bay

7 A 14 by 12 hangar — area

8 That hangar split at 10:
the 10 by 12 part

9 And the 4 by 12 part

10 Do the two parts add to
the whole — type the total

11 A 23 by 14 hangar —
area

12 24×15

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 36×14

.....

14. 45×16

.....

15. 53×12

.....

16. 28×25

.....

17. A hangar of area 144 with equal sides — one side

.....

18. Double both sides of a 14 by 12 hangar. Area multiplies by

.....

19. 10×4

.....

20. 20×6

.....

21. 30×5

.....

22. 40×4

.....

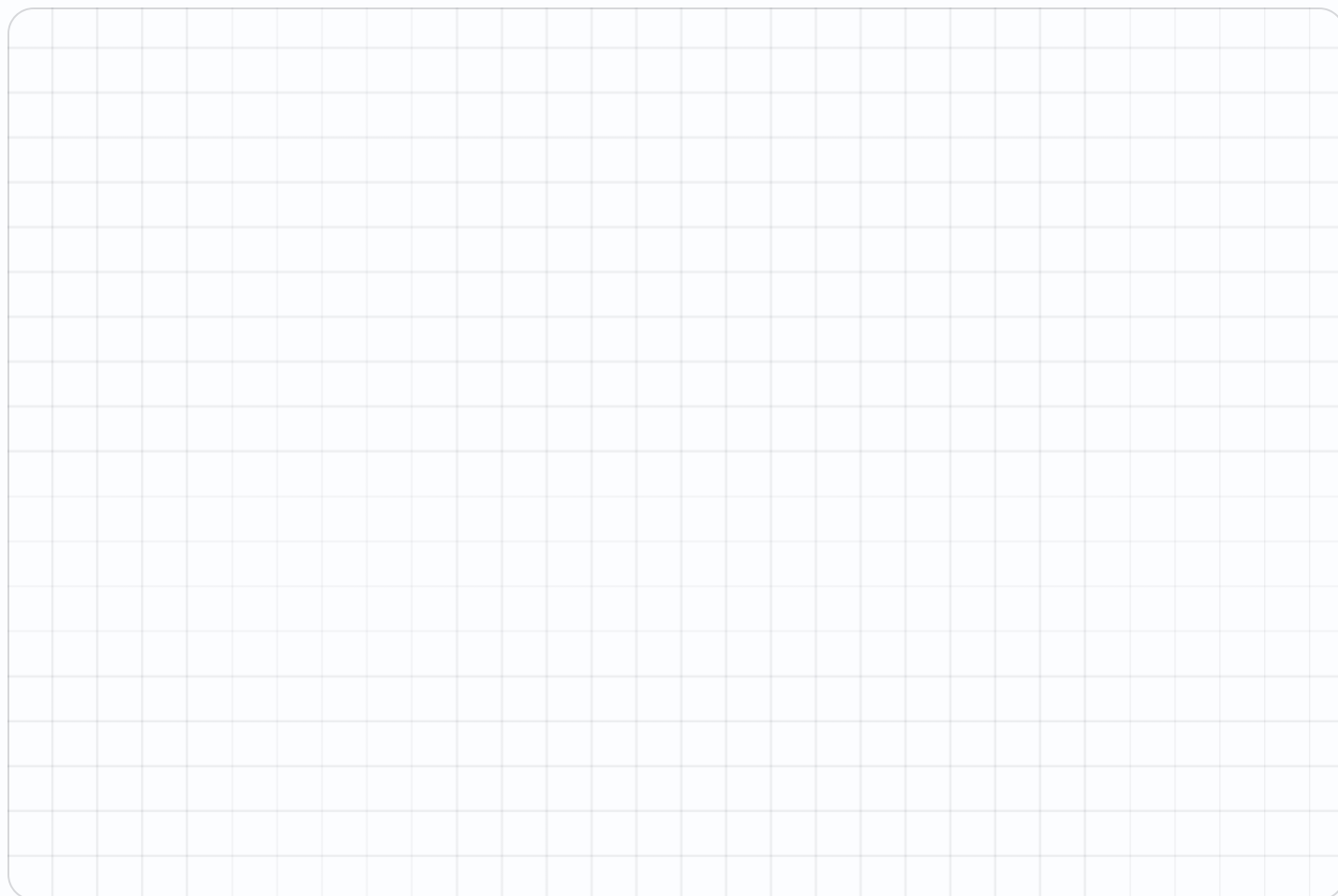
★ Challenge NOT GRADED • REASON IT OUT

23. 50×6
.....

24. 70×3
.....

25. 96×45
.....

26. 84×63
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Blueprint Defence

Both methods must agree, and you have to say why they do.

Warm-Up 11 ITEMS • RECALL

1 20×10

2 3×4

3 Sum of those four rooms

4 23×14 by algorithm

5 5×5

6 6×7

7 8×2

8 4×9

9 7×7

10 36×25 by area model
— the total

11 36×25 by algorithm

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12. The tensxtens room of 36×25

13. The onesxones room of 36×25

14. 47×18

15. 19×12

16. 28×13

17. 37×14

18. 46×15

19. 52×17

20. 20×4

21. 3×10

Error Journal Sweep

Mixed review of the whole mission. Fix only what repeats.

Warm-Up 12 ITEMS • RECALL

1 8×7

2 Is 9 prime — yes or no

3 Factors of 16 — how many

4 Third multiple of 7

5 2×5

6 4×3

7 6×8

8 9×9

9 7×8

10 38×24

11 GCF of 24 and 36

12 Largest prime factor of 84

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 23×16

.....

14. 41×18

.....

15. 35×21

.....

16. 29×24

.....

17. 48×19

.....

18. 6×40

.....

19. 5×18

.....

20. 9×26

.....

21. Estimate 296×5

.....

22. Primes below 100 — how many

.....



Working space



Error journal

Mission 01 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 7×9

2 8×7

3 6×9

4 23×14

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 45×23

6. Factor pairs of 36 — how many
_____7. GCF of 16 and 40
_____8. Is 51 prime — yes or no
_____9. Largest prime factor of 60

10. 32×14

11. 27×23

12. 8×30

13. 6×47

14. 326×4



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

5.1 • Missing-Digit Puzzles

OUT OF 15

Warm-Up • 1-10 • 6 | 8 | 9 | 7 | 5 | 8 | 24 | 24 | 20 | 27

Core • 11-15 • 3 | 23 | 6 | 8 | 13

Challenge • 16-26 • 32 | 1 | 70 | 50 | 168 | 204 | 235 | 392 | 342 | 1944 | 2856

5.2 • Launch Bay Blueprint

OUT OF 16

Warm-Up • 1-6 • 80 | 60 | 36 | 36 | 60 | 24

Core • 7-16 • 168 | 120 | 48 | 168 | 322 | 360 | 504 | 720 | 636 | 700

Challenge • 17-26 • 12 | 4 | 40 | 120 | 150 | 160 | 300 | 210 | 4320 | 5292

5.3 • Blueprint Defence

OUT OF 19

Warm-Up • 1-9 • 200 | 12 | 322 | 322 | 25 | 42 | 16 | 36 | 49

Core • 10-19 • 900 | 900 | 600 | 30 | 846 | 228 | 364 | 518 | 690 | 884

Challenge • 20-25 • 80 | 30 | 6 | 2852 | 2000 | 1728

Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • 56 | 240 | no | 5 | 90 | 21 | 10 | 12 | 48 | 81 | 56 | 50

Core • 13-22 • 234 | 912 | 12 | 7 | 1500 | 368 | 738 | 735 | 696 | 912

Challenge • 23-26 • 25 | 5568 | 3618 | 6000

Fri • Mission 01 Test

OUT OF 14

Warm-Up • 1-4 • 63 | 240 | 56 | 54

Core • 5-14 • 282 | 322 | 1035 | 1304 | 5 | 8 | no | 5 | 448 | 621

Challenge • 15-19 • 14 | 2356 | 3960 | 4000 | 8991